MIL-T-13133C
17 March 1981
SUPERSEDING
MIL-T-13133B(EL)
3 December 1965

MILITARY SPECIFICATION

THERMOMETERS, SELF-INDICATING, LIQUID-IN-GLASS ML-438( )/PMQ-1, ML-439( )/PMQ-1, ML-440( )/PMQ-1, AND ML-488( )/PM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the following items:

Thermometer, Self-Indicating, Liquid-in-Glass ML-438( )/PMQ-1
Thermometer, Self-Indicating, Liquid-in-Glass ML-439( )/PMQ-1
Thermometer, Self-Indicating, Liquid-in-Glass ML-440( )/PMQ-1
Thermometer, Self-Indicating, Liquid-in-Glass ML-488( )/PM

2 APPLICABLE DOCUMENTS

2.1 Issues of documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (i) specific issues are set forth therefor in the cited Specifications or (ii) different issues than those specified in the cited Specifications are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-13231	Marking of Electronic Items
MIL-P-14072	Finishes for Ground Signal Equipment
MIL-E-55585	Electronic Equipment and Parts, Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, NJ 07703, by using the self-addressed, Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6660

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-726	Packaging Requirements Code

DRAWINGS

ELECTRONICS COMMAND

SC-DL-87856	Thermometer ML-438( )/PMQ-1
SC-DL-87859	Thermometer ML-439( )/PMQ-1
SC-DL-87862	Thermometer ML-440( )/PMQ-1
DL-5072041	Thermometer ML-488( )/PM

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. This equipment consists of temperate, tropical, and arctic thermometers used in Psychrometer ML-436( )/PMQ-1 and a surface thermometer. The thermometers for use with the psychrometer shall permit temperature measurements over the range of -75°F to +140°F. The surface thermometer shall measure over the range of -50°F to +150°F.

3.2 First article. When specified the contractor shall furnish sample units, as required by contract, for first article inspection and approval (see 4.3 and 6.2d).

3.2.1 Inspection Standards. Concurrently with or prior to the submission of the first article units if required by the contract, the contractor shall submit a certificate of accuracy from some recognized standardizing agency covering the test standards, which he proposes to use in the graduation and inspection of instruments on order. The statement shall include a complete description of the standards, and the most recent test data taken to determine their accuracy. Full details of dates, methods used, and conditions under which data was taken, shall be given. Written approval shall be obtained before using these standards for calibration or testing the thermometers furnished on order. The following types of standards, certified at the times indicated, are acceptable:

1. Mueller Bridge and Platinum Resistance Thermometer	Check once a year for first 3 years, once every 3 years thereafter

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2. Mercury-in-glass Check once a year

3.3 Construction of the thermometers.

3.3.1 Psychrometric thermometers. The four psychrometric thermometers, ML-438( )/PMQ-1, ML-439( )/PMQ-1, ML-440( )/PMQ-1 and ML-488( )/PM shall be constructed in accordance with drawings SC-DL-87856, SC-DL-87859, SC-DL-87862 and DL-5072041 respectively, and the applicable paragraphs of this specification.

3.3.2 Tubes. The tubes of the psychrometric thermometers shall consist of a cylindrical glass stem with a round capillary bore and a glass bulb sealed on the end of the stem. The dimensions and construction shall conform to Drawings SC-DL-87856, SC-DL-87859, SC-DL-87862 and DL-5072041 for thermometers ML-438( )/PMQ-1, ML-439( )/PMQ-1, ML-440( )/PMQ-1 and ML-488( )/PM respectively.

3.3.3 Bulb. The bulb of all four thermometers shall be made of Corning Normal Glass, or equal. It shall be cylindrical in shape and smoothly and neatly joined to the lower end of the stem and shall act as a reservoir for the liquid. It shall be sensibly concentric with the stem. The walls of the bulb shall be blown as thin as possible consistent with necessary strength in order to reduce the time lag of the thermometer to a minimum.

3.3.4 Expansion chamber. An expansion chamber shall be provided at the top of the tube. This expansion chamber shall be sufficiently large to permit the thermometer to be exposed to a temperature of 160 degrees Fahrenheit without breakage. The shape of the expansion chamber shall be such that no breaking of the mercury column will result when the temperature decreases and the mercury goes back into the tube. The change in curvature of the interior surfaces of the expansion chamber, the shape of this space, and the smoothness of the interior surfaces shall be such that none of the liquid is retained in the expansion chamber when the temperature decreases sufficiently to empty it. The expansion space of the psychrometric thermometers shall be in the form of a tear drop, closely approximating that of the drawings referenced in 3.3.2 and such as to meet tests of 4.9. The top of the expansion chamber shall be 0.375 inch below the cap to insure visual inspection of the expansion space. The expansion space of the surface thermometer shall be a continuation of the round capillary bore.

3.3.5 Strain. There shall be no strains, other than in the vicinity of the juncture of the bulb and stem (4.9).

3.3.6 Stem. The stem shall be made of good quality thermometric glass of a type suitable for the range of temperature to be measured. The stem shall have a strip of white opalescent glass incorporated therein, running

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the full length of the stem. This strip shall be so positioned with respect to the scale graduations as to provide a background to facilitate reading.

3.3.7 Graduations. The scale graduations shall be etched on the stem, in one-half degree Fahrenheit divisions on the arctic thermometer ML-438( )/PMQ-1, in whole-degree Fahrenheit divisions on the temperate and tropical thermometers ML-440( )/PMQ-1 and ML-439( )/PMQ-1, respectively and in two-degrees Fahrenheit divisions on the surface thermometer ML-488( )/PM. The graduations shall be not less than 0.003 inch nor more than 0.006 inch wide. Three lengths of graduation lines shall be used on the arctic thermometers and two lengths on the surface, temperate and tropical thermometers. On the arctic thermometer, the zero and each succeeding five-degree and ten-degree line shall be sensibly longer than the intermediate whole-degree lines, which in turn shall be longer than the half-degree marks. The temperate and tropical thermometers shall have the zero and each five-degree and ten-degree lines sensibly longer than the intermediate degree lines. The surface thermometer shall have the zero and each ten-degree lines sensibly longer than the intermediate degree lines. All lines, figures, and letters shall be clear cut, distinct and shall be filled with black pigment which is resistant to temperature conditions and chemicals to which the thermometer will be exposed.

3.3.7.1 Tube marking. All markings shall be placed on the opalescent backing in such a manner that they may be easily read. On the three-psychrometric thermometers the zero and each fifth degree-mark above and below zero shall be numbered. On the surface thermometers the zero and each twentieth degree-mark above and below zero shall be numbered. When the thermometer is held with stem vertical, bulb at the bottom, and the opalescent backing away from the observer, the numbers shall be erect and in front of the bore: the graduation lines shall lie in a straight line perpendicular to the bore. If the thermometer is gas-filled, the name of the gas used in filling the space in the stem above the thermometric liquid (3.3.4) shall be etched on the stem followed by the word "filled," for example, "Neon Filled." The letters composing those words shall be as small as requirements of etching and readability will permit. They shall be placed where they will not interfere in any way with the scale markings. The words "Mercury-Thallium" or "Mercury" as applicable shall also be etched on the stem in an adjacent area.

3.3.8 Provisions for wick. The outer diameter of the stem of the three psychrometric thermometers shall be decreased slightly at a distance of 0.125 inch above the junction of the stem and bulb in order that a psychrometric wick may be attached to the thermometer.

3.3.9 Scale designations. Suitable identification for Fahrenheit shall be etched on the stem in a prominent place.

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3.3.10 Range and accuracy. The range and accuracy of the thermometers shall be as shown in the following table. (See 4.6).

TABLE I. Range and accuracy.

Thermometer Type	TEMPERATE RANGE		Accuracy (°F.)
	Low(°F.)	High(°F.)	
ARCTIC (ML-438( )/PMQ-1)	-75°(+5° or -0°)	-25°(±5°)	±0.4°
TEMPERATE (ML-440( )/PMQ-1)	-35°(±3°)	60°(±5°)	±0.4°(-35° to 0°) ±0.2°(0° to 60°)
TROPICAL (ML-439( )/PMQ-1)	40°(±5°)	140°(±5°)	±0.2°
SURFACE (ML-488( )/PM)	-50°(±5°)	150°(±5°)	±0.5°

3.3.11 Calibration points. The contractor shall furnish a statement identifying ten calibration points equally (approximately) distributed throughout the temperature range of each thermometer. (See 4.6.2).

3.4 Parts, materials, and processes.

3.4.1 Glass. All glass shall be thoroughly annealed or seasoned in accordance with the best accepted practice in the manufacture of meteorological thermometers.

3.4.2 Thermometric liquid. The mercury shall be triple-distilled and chemically pure. The mercury surface shall be clear and have a brilliant appearance. There shall be no separation of the mercury in the thermometer tube due to improperly cleaned mercury (4.10). The space above the mercury shall be either evacuated or filled with an inert gas such as Argon, Krypton, Neon, etc., and if gas-filled shall show no visible evidence of oxidation. The thermometer-filling method and pressures employed shall conform to the best practice in thermometry.

3.4.2.1 Tropical and temperate range. The thermometric liquid used in the tropical and temperate range thermometers shall be chemically pure mercury as specified in 3.4.2.

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3.4.2.2 Arctic and surface range. The thermometric liquid used in arctic and surface range thermometers shall be mercury and thallium amalgam having a freezing point not higher than -76°F. The mercury-thallium shall be in such proportions and purity that the thermometers will satisfactorily meet the specified requirements and tests (see 4.10 and 4.11).

3.5 Finish, protective. Each thermometer cap shall be given protective finish in accordance with Specification MIL-F-14072.

3.6 Marking. Marking shall conform to Specification MIL-M-13231, except that any group I, II or III marking and any material and process for labels, as described in that specification, may be used for surfaces exposed to view during use or storage of the equipment (see 4.4.1).

3.7 Service conditions. The thermometers listed in 1.1 shall meet the following service conditions:

3.7.1 Thermometer operating. Except as otherwise specified herein, all thermometers listed in 1.1 shall meet the requirements of this specification under any probable combination of the following:

a. Operation. Operation of the thermometers in any specific manner.
b. Temperature. The thermometers shall operate throughout their respective calibrated ranges.

3.7.2 Thermometer non-operating. Except as otherwise specified, all items specified in 1.1 shall meet the requirements of 3.7.1 after subjection to any one or more of the following conditions:

a. Temperature. Continuous exposure for 72 hours at 160°F and 72 hours at -80°F, in any orientation.
b. Orientation. Indefinite storage in any orientation.

3.8 Workmanship. The thermometers shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of the following paragraphs:

3.3.3	Bulb	3.3.8	Provisions for wick
3.3.4	Expansion chamber	3.3.9	Scale designations
3.3.6	Stem	3.4.2	Thermometric liquid
3.3 7	Graduations	3.5	Finish
3.3.7.1	Tube marking	3.6	Marking

3.9 Maintenance. The thermometers shall require no maintenance other than cleaning on an "as required basis."

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3.10 Human factors. The thermometers shall be capable of being read while held in one hand by a standard observer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Packaging inspection (see 4.12).

4.3 First article inspection. This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified for group A and group B (see tables II and III respectively).

4.4 Quality conformance inspection. The contractor shall perform the inspection specified in 4.4.1 through 4.4.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.4). Test equipment for Government verification inspection shall be made available by the contractor.

4.4.1 Group A inspection. This inspection including sampling, shall conform to table II and inspection procedures of standard MIL-STD-105, using the general inspection levels. Group A order of inspection shall be at the discretion of the contractor, except that the test of expansion space (see 4.8) shall be last.

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TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test Method paragraph	AQL (percent defective)	
			Major	Minor
Visual and dimensional				
Construction	3.3, 3.6 3.7, 3.8	4.4.1, 4.5	1%	4%
Mechanical	3.8	4.5	1% (for group)	
Accuracy	3.3.10	4.6		
Liquid Break (Mercury or mercury-thallium)	3.4.2	4.7		
Expansion chamber	3.3.4	4.8		

4.4.2 Group B inspection. This inspection, including sampling, shall conform to table III and procedures of MIL-STD-105, using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S-4 for normal, tightened and reduced inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Inspection paragraph
Strain	3.3.5	4.9
Purity (Mercury or alloy)	3.4.2	4.10
Freezing Point (Mercury-thallium alloy)	3.4.2.2	4.11

4.5 Visual and mechanical inspection. Thermometers shall be examined for defects listed in table IV.

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TABLE IV. Classification of visual and mechanical defects.

Classification	Defects
Major	Etching not continuous and uniform, poorly defined. Missing graduation, incorrect position, or numbering. Markings missing, wrong, or incomplete. Pits, bubbles, cracks, or distortion in glass. Broken or damaged part affecting use. Dimensions out of tolerance affecting use. Deformed bulb or tube affecting use for which intended. Stem strip out of position, does not provide proper background to facilitate reading.
Minor	Dimensions out of tolerance, not affecting use or interchangeability. Slight deformities not affecting use for which intended. Marking and surface defects affecting appearance but but not use.

4.6 Test for accuracy.

4.6.1 Test conditions In performing the test for accuracy, the thermometers shall be tested under the following conditions:

4.6.1.1 The thermometers to be tested shall be immersed in a fluid bath (provided by the contractor) which shall be thoroughly stirred so that any temperature difference between the measuring equipment and the instruments under test do not exceed 0.03°F. The rate of drift of the temperature of the bath shall be no more than 0.02°F per minute. The bath shall be capable of providing temperatures covering the over-all range. (3.3.10).

4.6.1.2 An appropriate working standard thermometer for measuring the true temperature of the bath, shall be immersed in the same bath as the thermometers being checked, and centrally located with respect to them. Both the working standard and the test thermometers shall have their bulbs placed at approximately the same level in the bath. The standards shall cover the full operating range of the thermometers and shall have accuracies of ±0.05°F throughout the over-all range covered by the thermometer under test.

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4.6.1.3 All thermometers being tested shall be totally immersed in the bath or shall have their stems immersed to within 0.25 inch of the top of the mercury column.

4.6.1.4 The temperature of the liquid bath shall be held constant to within ±0.03°F during the time test data are taken. Data shall not be taken until both the working standard thermometers and those being tested have completely responded to the temperature of the bath.

4.6.2 Test. The temperature indicated by the thermometer being tested shall be compared with the true temperature of the thermometer bath (4.6.1.1) as indicated by the working standard thermometer (4.6.1.2). This comparison shall be made at ten points approximately equally distributed throughout the range of the scale. The difference between true temperature and the temperature indicated by the thermometer being tested shall not be more than the accuracy specified (table I). All test temperatures shall be selected at points intermediate between the manufacturer's calibration points.

4.7 Breaks in mercury-thallium or mercury. The thermometer shall be inspected for breaks in the mercury-thallium or mercury column. If no break is found, one shall be produced artificially, if possible, by tapping the thermometer lightly while it is held in a horizontal position. The tapping shall be vigorous enough to simulate actual shipping conditions. If such breaks are found or created, the breaks shall be capable of being corrected using manual corrective measures with no resulting damage to the thermometer. (See 3.4.2).

4.8 Test of expansion space. Thermometers of each range shall be tested for one minute in a water bath maintained at 160° ± 2°F. The thermometers shall then meet the requirements of 3.3.4.

4.9 Test for strain. Thermometers shall have their tubes examined for strain by means of a polariscope. There shall be no evidence of strain at points other than in the vicinity of the juncture of the bulb and stem. There shall be no abrupt change in color at any other point. (3.3.5).

4.10 Purity test of mercury or alloy. An inspection of the thermometric liquids shall be made to determine compliance with requirements of 3.4.2. Test certificates or other adequate evidence of the chemical purity of the mercury or mercury and thallium used shall be submitted to the contracting officer.

4.11 Test for freezing point. The contractor shall demonstrate to the inspector's satisfaction that mercury-thallium alloy used in the thermometers meets the freezing point requirements of 3.4.2.2.

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4.12 Quality Conformance Inspection of Section 5. Quality conformance inspection of Section 5 shall be in accordance with group A, group B, and when required (see 6.2g) for group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 1Ø-1-1-ØØ-LA-BG-X-CT-Q-ED-A.

5.1.2 Level B. 1Ø-1-1-ØØ-LA-BG-X-CT-Q-ED-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The thermometers covered by this specification are to be used in Psychrometer ML-436( )/PMQ-1 as wet and dry bulb thermometers for measuring the difference in temperatures of the atmosphere constituting a measure of the humidity.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.
b. Type required.
c. Level of packaging (level A or level B).
d. Samples required (see 3.2).
e. Marking and shipping of samples.
f. Place of final inspection.
g. Whether or not rough handling tests are required.

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: ML-439W/PMQ-1. As soon as possible after award of the contract, the contractor should apply to the Government office specified in the contract for such information (see 1.1)

6.4 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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